



Chitosan macroporous asymmetric membranes—Preparation, characterization and transport of drugs

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ABSTRACT

Chitosan macroporous membranes with asymmetric morphology were obtained by using an inorganic porogen agent (SiO_2). Chitosan/silica ratios used were 1:1, 1:3, and 1:5 w/w. A methodology to obtain asymmetric membranes with control of porosity and average pore size was proposed. The porous membranes were obtained taking advantage of the opposite solubility characteristics of chitosan and silica (4–20 μm). The membranes were characterized by SEM and water sorption capacity. The porosity was calculated by the relationship between dense and macroporous membranes. The SEM images of both surfaces and cross-section of the membranes confirmed their asymmetric morphology. Using a double-cell method, the permeability coefficients of two model drugs (sodium sulfamerazine and sulfamethoxypyridazine) were determined. The effects of porous layer, drug type, concentration and temperature were evaluated. The results revealed that the increase in porosity results in significant differences in permeability and that the effects of drug concentration and bath temperature become less pronounced as porosity increases. The mass transport was analyzed in terms of pore-flow mechanism and the solution-diffusion mechanism. The results showed that the methodology was very efficient to yield asymmetric membranes with good mechanical resistance, control of porous size and dense layer thickness and that these membranes can potentially be used to the transport of drugs.

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1. Introduction

In recent years, many researchers have been directed toward the development of improved transdermal, nasal and oral drug delivery systems, scaffolds and wound dressing using biodegradable materials. Chitosan is a partially deacetylated derivative of chitin, a by product of the seafood industry and, in view of its hydrophilicity, excellent film-forming capacity, good mechanical properties, high chemical reactivity and availability, it can be an excellent candidate for such systems [1–8]. In many cases of such applications a porous structure is desirable. Techniques for attainment of macroporous chitosan membranes have been suggested and a variety of asymmetric and symmetric structures have been obtained [9–15]. For symmetric structures, porogen agents have showed good results in yield structures with easy control of average pore size. The technique is based on inverse porogen/polymer solubility and consists on dissolution of the porogen agent in the polymeric matrix, followed by solvent evaporation and porogen agent extraction by immersion on a suitable solvent. Zheg and Ruckenstein employed such a technique to prepare macroporous chitosan membranes

using silica as porogen agent and obtained symmetric membranes that have emerged as an alternative to affinity column chromatography, and protein separations [9,11]. The control of pore size is a function of the size of the silica particles used, which represent an advantage for this kind of membrane. Many other applications have been related like adsorption of bacterial endotoxins [16], concanavalin [13] and model proteins like human serum albumin (HSA) and bovine serum albumin (BSA) [17]. Besides silica, many other compounds have been used as porogen agents like NaCl for chitosan [14] and CaCO_3 for chitin [18]. The main disadvantages of such membranes are their low mechanical resistance (usually improved by cross-linking) and low selectivity. In order to overcome these limitations the use of asymmetric membranes has been proposed. The asymmetric membrane has a dense layer and a sponge like sub-layer and has been suggested as a quite suitable material for wound dressing applications since the dense skin layer would be able to prevent bacterial invasion and possibly to act as rate-controlling layer for water vapor permeation, while the inner layer is designed for drainage of wound exudates [12]. The recommended fabrication method is the phase inversion method. In summary, the chitosan solution is prepared, pre-evaporated, cast and then immersed into a coagulation bath for long periods of time. When the solvent out-flow rate is larger than the coagulation in-flow, a dense structure is formed. As the ratio is smaller, the casting solution separates

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into polymer-rich and polymer-poor phase, the former results in polymer matrices and the latter forms the porous layer. By controlling the phase separation process (especially by evaporation time and temperature), different structures can be fabricated. Asymmetric membranes with high flow rate and good permselectivity, have also been currently used for gas and vapor separation [8]. In this work one proposes a new pathway that tries to combine the best of the two above-mentioned methodologies: the preparation of asymmetric membranes with a dense layer that provides good mechanical resistance and selectivity, and the use of a porogen agent to produce a porous layer with a good control of pore size. The methodology also allows the control of the dense layer thickness that is important for the final membrane properties such as permeability.

2. Materials and methods

Chitosan was supplied by Polymar Ltd. (Brazil) with a degree of deacetylation of 85%, according to the supplier. Silica gel was purchased from Sigma (USA) with a particle size distribution between 4 and 20 μm . Sulfamerazine, and sulfamethoxypyridazine, both R&D grade, were purchased from Sigma (USA). All others chemicals were analytical grade reagents and used as received.

2.1. Membrane preparation

2.1.1. Dense membranes

Chitosan was dissolved in a 2.0% aqueous acetic acid solution for 24 h under mechanical stirring in order to form a 1.5% solution. The resultant solution was filtered using a Millipore® Millex filter. Following, the solution rested for 2 h to allow bubbles evolution. Thereafter, part of solution was cast on a glass plate and dried in oven at 50 °C for 3 h. Different amounts of silica dispersion were added to the remaining solution under vigorous stirring, using chitosan/silica ratios of 1:2, 1:6, and 1:10. After 3 h the dispersions were poured into the glass plate, which has been already in oven, with a 1:1 ratio and left for more 24 h in oven until complete solvent evaporation. With that procedure, chitosan/silica final ratios obtained were 1:1, 1:3 and 1:5.

Pure chitosan membranes were obtained using the same initial chitosan solution concentration. In this case, the chitosan solution was cast on the glass plate and left in oven for 24 h for solvent evaporation. After that, the membranes were immersed in a 0.5 M sodium hydroxide (NaOH) solution for 2 h, to be neutralized. Following, they were repeatedly washed with distilled water and allowed to dry at room temperature. Such membranes were named CS membranes.

2.1.2. Macroporous membranes

The macroporous membranes were obtained by treatment of the dense membranes with a sodium hydroxide (NaOH) 5% solution at 60 °C for 4 h in order to dissolve the silica particles. Thereafter, the membranes were washed with distilled water, firstly at 60 °C and subsequently at room temperature to completely remove the alkali excess. The temperature of 60 °C was chosen in order to preserve the membrane's physical and chemical properties. No changes in color were observed at this temperature. After that, the membranes were immersed in a 20% glycerol solution for 30 min, washed with distilled water and allowed to dry at room temperature. The role of glycerol was to plasticize the membranes, increasing their processability and to avoid the closure of the porous during the drying process. The macroporous membranes were named MPCSI1, MPCSI13 and MPCSI15 for chitosan/silica ratios of 1:1, 1:3 and 1:5, respectively.

2.2. Characterization

2.2.1. Membrane thickness, porous size and porous layer

The dry membrane thickness was measured with a digital micrometer Digiderm, Mitutoyo. Each membrane was measured at 10 different points (5 points each side) and only the membranes with thickness standard deviation smaller than 10% were used. The porous layer percent was estimated from the SEM cross-section results using the UTHSCSA program-image tool for windows version 3.00. In this case, 40 measurements at different points were done and the average and standard deviation were obtained. The program was also used to calculate the average pore size distribution using the same number of measurements.

2.2.2. Membrane porosity

The sample porosity (ε) was calculated according to

$$\varepsilon = \frac{1 - \rho_d}{\rho_p} 100 \quad (1)$$

where ρ_p and ρ_d are the macroporous membrane density and dense membrane density, respectively. ρ_p was calculated by the relation between the porous membrane dry weight (W) and volume (V). The porous membrane dry weight was obtained by successive drying procedures in oven until constant weight. The volume of membrane (V_m) was calculated according to

$$V_m = \frac{1}{4} \pi d^2 l \quad (2)$$

where d is the membrane diameter and l the membrane thickness. Five samples were used for each silica proportion used in this work. ρ_d was calculated using the same methodology, replacing the porous membranes by the dense chitosan membrane. In this case, eight samples were used.

2.2.3. Scanning electron microscopy

Sample preparation changed according to the kind of image desired, surface or cross-section. For surface images, a small piece of membrane (dense and porous) was cut and sputtered coated with a thin gold layer before scanning. For the cross-section image, the membrane was freeze-fractured under liquid nitrogen in order to obtain a sharp brittle fracture surface. Following, the membrane was sputtered coated with a thin gold layer. The images were obtained by a SEM Philips XL30 (Netherlands) using a tungsten filament, secondary electrons detector and a voltage of 20 kV.

2.3. Water sorption capacity

The water sorption capacity of membranes was estimated through a mass balance (Mettler H35AR) at room temperature. The dry membranes were weighted and then immersed in distilled water. At time intervals of 30 min, 1, 2, 3, 4, 5, 6, 24 and 48 h the membranes were removed from water, had the water from surface wiped off, weighted and returned to water. The water sorption capacity ($W\%$) was calculated according to

$$W (\%) = \frac{m_w - m_d}{m_d} 100 \quad (3)$$

where m_w and m_d are the mass of wet and dry membrane, respectively. In these experiments, four samples were analyzed for each type of membrane.

2.4. Drug permeability test

The permeability coefficients of two model drugs were determined using a permeability cell made of two plastic compartments

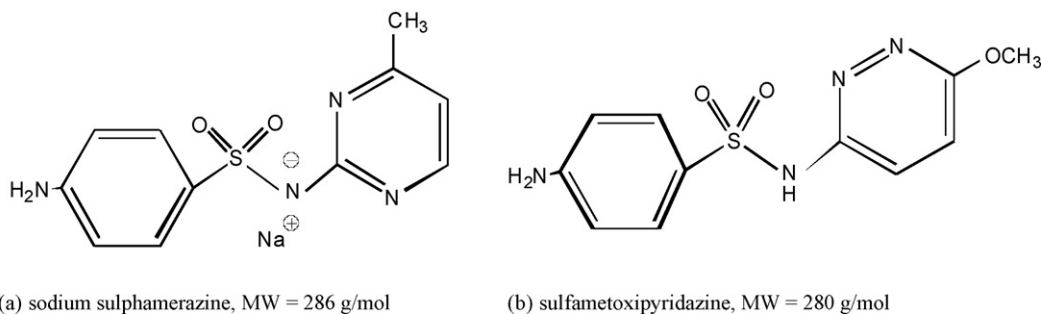


Fig. 1. Chemical structure for (a) sodium sulfamerazine, MW = 286 g/mol and (b) sulfametoxypyridazine, MW = 280 g/mol.

with 230 ml of capacity already described elsewhere [19,20]. The membrane area was 8.465 cm². Solutions inside the cell compartments were mechanically stirred. The drugs used in the experiments were sodium sulfamerazine and sulfametoxypyridazine which structures are shown in Fig. 1. Prior to the experiment the membranes were swollen in water for 12 h until an equilibrium state was reached. The right side of the cell was filled with a drug solution with concentration of 0.1%, 0.3%, 0.5%, 1.0% and 1.5% for sodium sulfamerazine and 0.1% for sulfametoxypyridazine. The left side of the cell was always filled with distilled water. During the experiments the right side concentration was considered constant. A continuous flux system was attached to the left

side of the cell in order to determine the solution absorbance at $\lambda = 260$ nm for sodium sulfamerazine and $\lambda = 261$ nm for sulfametoxypyridazine, using a UV-vis spectrometer (UV Genesis 10). The temperature was maintained by thermostatic bath at 30 ± 0.1 , 35 ± 0.1 and 40 ± 0.1 °C. All experiments were run in duplicate.

The permeability coefficients were calculated using the model described by Crank for flux through a membrane [21] adapted to measurements of absorbance values [20]. In this case, permeability coefficients can be calculated according to

$$P = \frac{\alpha V I}{C_1 \epsilon b S} \quad (4)$$

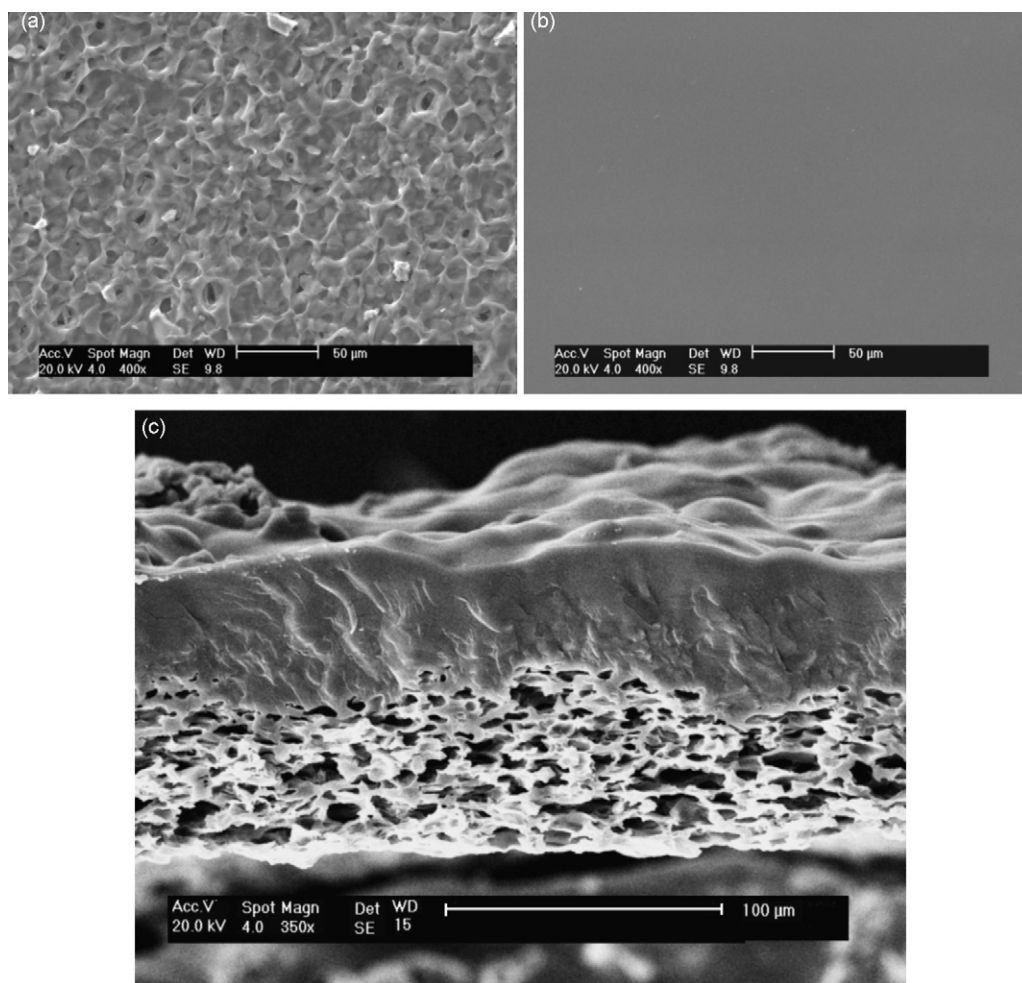


Fig. 2. SEM micrographs for MPC511 membrane: (a) bottom surface; (b) top surface and (c) cross-section.

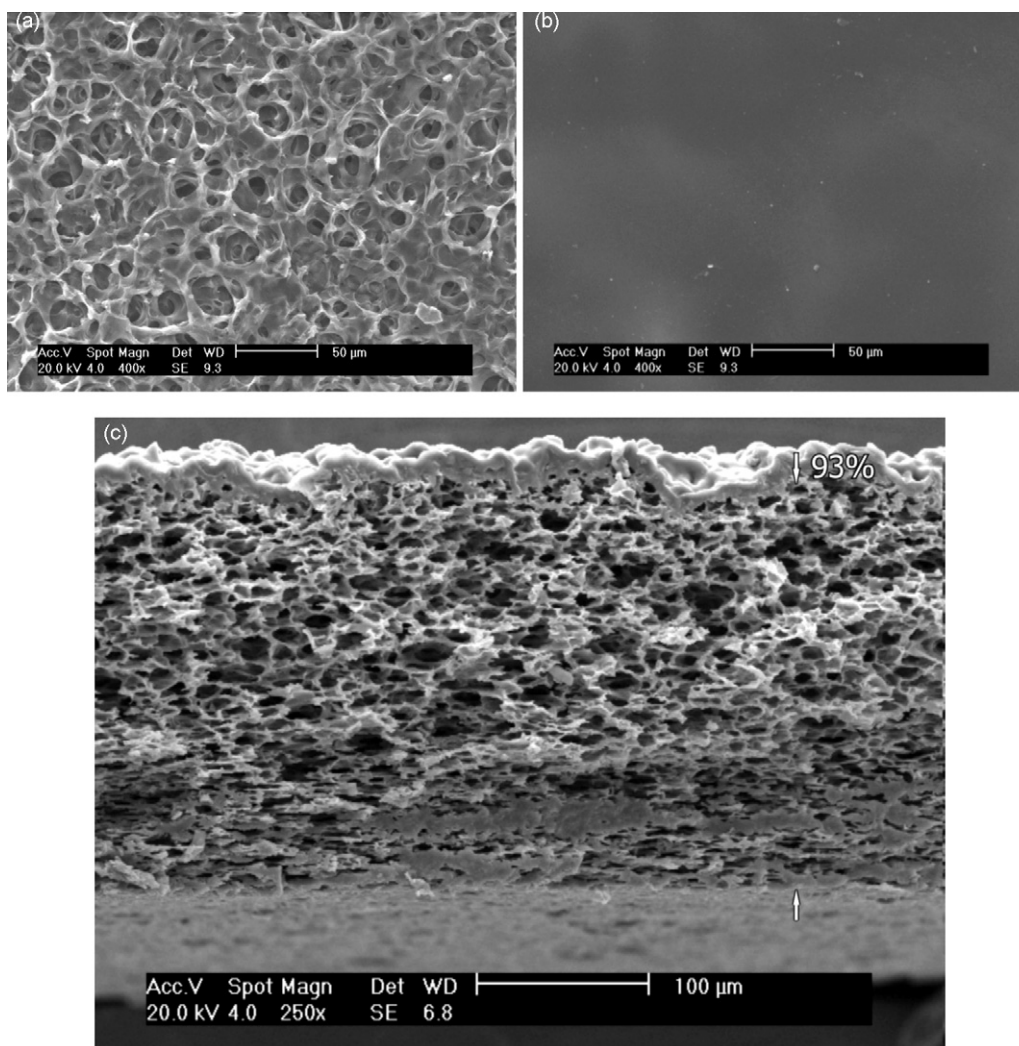


Fig. 3. SEM micrographs MPCS13 membrane: (a) bottom surface; (b) top surface and (c) cross-section.

where α is the angular coefficient of the curve generated by the plot of absorbance versus time, V is the cell volume, l membrane thickness, C_1 initial drug concentration, ε absorptivity, b the UV–vis cell pathway and S the membrane area.

3. Results and discussion

3.1. Membrane thickness, porosity, pore size and porous layer

The results for membrane thickness, porosity pores size and porous layer are presented in Table 1. The results show that membrane thickness increases with increases in the amount of silica used. Considering that the amount of chitosan used in each membrane is exactly the same for all compositions prepared, one can conclude that such increase in thickness is consequence of the presence of pores that increase the membrane total volume. The increments in porosity and porous layer are also proportional to the amount of silica added. It is also possible to observe that for MPCS15 membrane the porous layer value is very close of its maximum which means that higher amounts of silica, although increase the membrane's porosity, will cause an asymmetry loss.

3.2. Scanning electron microscopy

The morphology of macroporous chitosan membranes using silica ratios of 1:1, 1:3 and 1:5 can be seen from the micrographs showed in Figs. 2–4, respectively. During the drying process, one membrane's surface was in contact with the glass plate while the other was in contact with air. From now on, these surfaces will be named bottom and top surfaces, respectively. Figs. 2–4 show images of the bottom and top surfaces, as well as the cross-sections for the

Table 1

Values of membrane thickness, porosity, pore size and porous layer for dense chitosan membrane (CS), macroporous chitosan membranes with 1:1 chitosan:silica ratio (MPCS11), 1:3 chitosan:silica ratio (MPCS13) and 1:5 chitosan:silica ratio (MPCS15)

Membrane	Thickness (μm)	Porosity (%) ^a	Pore size (μm) ^b	Porous layer (%) ^b
CS	49 $\pm$ 1	18 $\pm$ 2	17 $\pm$ 4	$\approx$ 55
MPCS11	69 $\pm$ 5	18 $\pm$ 2	17 $\pm$ 4	$\approx$ 55
MPCS13	180 $\pm$ 10	55 $\pm$ 3	22 $\pm$ 4	$\approx$ 95
MPCS15	260 $\pm$ 10	63 $\pm$ 2	32 $\pm$ 5	$\approx$ 100

^a From Eq. (1).

^b Determined by SEM.

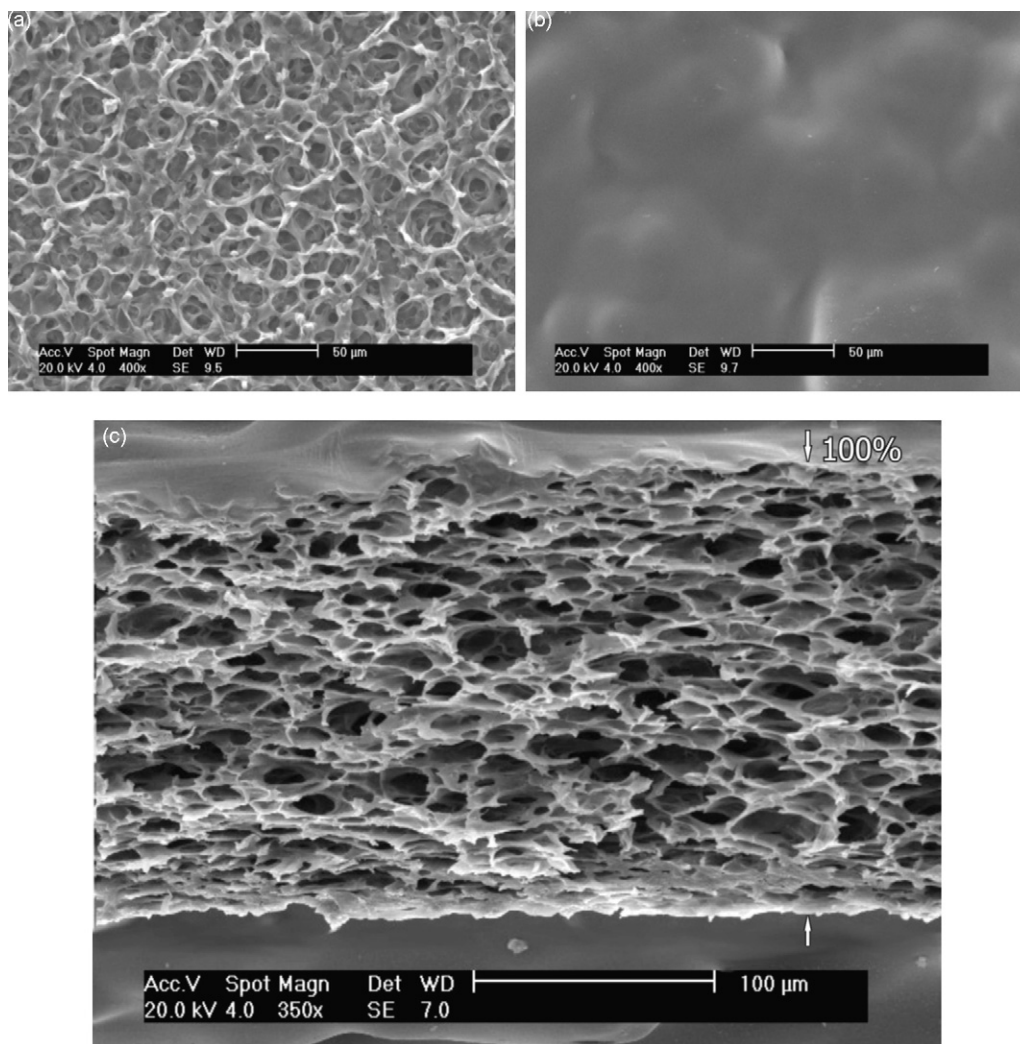


Fig. 4. SEM micrographs for MPCS15: (a) bottom surface; (b) top surface and (c) cross-section.

porous membranes. For the three macroporous samples analyzed (MPCS11, MPCS13 and MPCS15) it can be observed that the porous formation, due to silica dissolution in alkaline medium, is always evidenced at the bottom surface. Initially it was supposed that the proposed methodology would form asymmetric membranes where the porous layer would be formed at the top surface. However, the results showed that silica particles present on the dispersion were able to cross over the pure chitosan film (partially dry) being deposited at the bottom of the glass plate. Based on these results, the process occurring upon fabrication of the asymmetric macroporous membranes was schematically described according to Fig. 5.

Comparing Fig. 2(a), 3(a) and 4(a) one can state that the pores are regularly distributed and that their density increases with the increase in the initial silica content. It can also be observed that the average pore size is approximately in the same range of the silica particle diameter (between 4 and 20 µm), indicating that the pores' size can be controlled by selecting silica particles of appropriated size. Comparing the bottom and top surfaces for the three compositions analyzed (parts (a) and (b) of Figs. 2–4) one can confirm the membranes' asymmetry, since part (b) showed no evidence of pores, even for the MPCS15 membrane. In other words, while the bottom surface, that was rich in silica, was completely modified by the NaOH treatment, generating the porous surface, the top surface composed only by chitosan was not modified by the presence

of alkali, maintaining its dense morphology. In part (c) of Figs. 2–4 the membranes' cross-sections are shown. The images confirm the membranes' asymmetry. All membranes are formed by one surface and part of their volume containing pores while the other surface is dense and continuous. It is possible to observe, especially for MPCS15 membrane, the presence of some few pores with average size bigger than the expected. These pores could be a consequence of the collapse of some vicinal pores during the treatment with NaOH. This effect was confirmed through the statistical analysis of pore size, using the UTHSCSA program which produced a higher pore size for MPCS15 membrane (Table 1). The higher occurrence of collapse on MPCS15 membranes is a consequence of higher amount of the silica particles in such membrane. It can also be observed that the dense layer is on its limit, indicating that such composition is a limit for obtaining a membrane with asymmetric characteristics. Therefore, for the system under study, one can say that any amount of silica higher than the one used on MPCS15 membrane will result on a symmetric membrane with pores in both surfaces. It is worthwhile to remind that the asymmetry is an important factor since the dense layer, even when very thin, has two important functions: to confer mechanical resistance and selectivity to the membrane. Fig. 6(a)–(c) show the bottom surface, top surface and cross-section of a pure chitosan membrane, respectively, for the sake of comparison.

Table 2

Permeability coefficients for sodium sulfamerazine in CS, MPC11, MPC13 and MPC15 membranes at different drug concentrations and temperatures

	Permeability for sodium sulfamerazine— $P \times 10^6$ (cm ² /min)						
	$T = 30^\circ\text{C}$					$T = 35^\circ\text{C}$	$T = 40^\circ\text{C}$
	0.1 ^a	0.3 ^a	0.5 ^a	1.0 ^a	1.5 ^a	0.1 ^a	0.1 ^a
CS	9.3 ± 0.3	7.3 ± 0.2	6.7 ± 0.4	5.1 ± 0.9	4.9 ± 0.2	11.0 ± 0.1	12.6 ± 0.1
MPCS11	25.7 ± 0.7	23.8 ± 0.6	16.4 ± 0.5	12.9 ± 0.8	11.5 ± 0.2	39.5 ± 0.5	43.0 ± 0.4
MPCS13	37.2 ± 0.1	40 ± 2	32.5 ± 0.4	32.1 ± 0.8	32 ± 2	53.9 ± 0.1	64.4 ± 0.4
MPCS15	55 ± 3	50 ± 3	51 ± 3	50.3 ± 0.6	53 ± 4	57.6 ± 0.1	59.7 ± 0.4

^a Drug concentration (%).

3.3. Water sorption capacity

Fig. 7 shows the results of water sorption for CHI, MPC11, MPC13 and MPC15 membranes. The results show a dramatic increase in water sorption capacity with the increase in porosity indicating the water filled macroporous nature of these membranes. These results, added up to the SEM results, are important to elucidate the solute transport mechanism in such membranes. So far, one can suppose that the transport through the water filled channels will be dominant in this case. However, due to the presence of a dense layer composed by a polymer matrix, a possible secondary transport through the polymer might also be considered.

3.4. Drug permeability tests

A series of experimental results for permeation of sodium sulfamerazine on MPC11 membranes at different temperatures is

shown in Fig. 8 for illustrative purposes only. It can be seen that the absorbance versus time plots yield straight lines and present good reproducibility. The slope of these lines can be used to calculate the permeability coefficient, P as indicated in Eq. (4). The calculated values of P for different membranes (CS, MPC11, MPC13 and MPC15), drug concentrations (0.1%, 0.3%, 0.5%, 1.0%, and 1.5%) and temperatures (30, 35 and 40 °C) are summarized in Table 2 for sodium sulfamerazine and in Table 3 for sulfametoxyipyridazine. Starting the analysis from pure chitosan it is possible to observe that the more drug was loaded, the lower the permeability coefficient was. This result is very interesting, considering that as more drug was loaded, the cumulative release amount is expected to increase and therefore a more persistent release can be achieved by increasing the drug concentration. Similar results were obtained by Wang et al. when studying the permeability of ciprofloxacin hydrochloride in chitosan films [22]. The decrease on permeability with increase in concentration could indicate some possibilities: a decrease on drug diffusion coefficient or a decrease on the drug distribution coefficient between the liquid and chitosan phase. When the experimental curves (absorbance versus time) present a time lag, the diffusion coefficient can be easily calculated [23]. However, the experiments presented here showed no induction time and therefore the diffusion coefficient should be calculated from drug distribution coefficient values. Unfortunately, the results of drug distribution coefficients were not conclusive, probably due to a combination of low solute concentration and low membrane thickness. Another possibility is that the decrease in permeability is a consequence of the liquid diffusion boundary layers which form on each side of the membrane and which can control the drug diffusion at low concentrations.

3.4.1. Influence of membrane's structure

The macroporous membranes described here have a distinctive asymmetric structure with a dense polymer layer and a cellular structure with highly interconnected porosity. The results show that the increase in porosity results in an increase in permeability. The significant higher values of permeability on such macroporous hydrogel membranes highlight the importance of solute transport through the pores in the swollen gel (pore-flow mechanism). Falk et al. [24] have shown that increasing the volume fraction of macropores in chitosan hydrogel can lead to a significant improvement

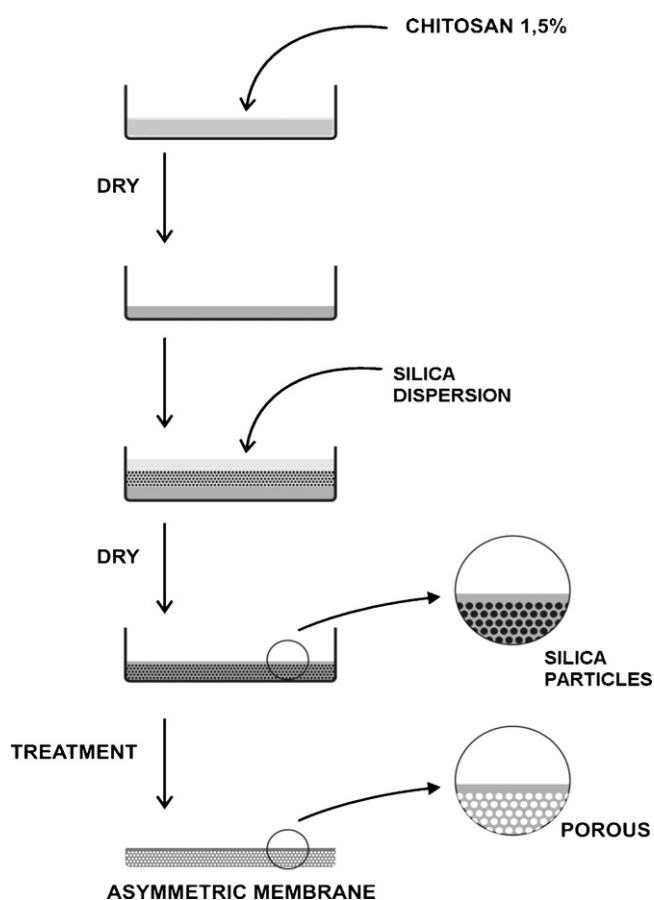


Fig. 5. Preparation of macroporous asymmetric chitosan (CS) membranes. Silica particles (full circles); pores (open circles).

Table 3

Permeability coefficients for sulfametoxyipyridazine (0.1%) in CS, MPC11, MPC13 and MPC15 membranes at 30 °C

	Permeability for sulfametoxyipyridazine 0.1%— $P \times 10^6$ (cm ² /min), $T = 30^\circ\text{C}$
CS	1.7 ± 0.4
MPCS11	5.2 ± 0.4
MPCS13	6.5 ± 0.5
MPCS15	18.5 ± 0.5

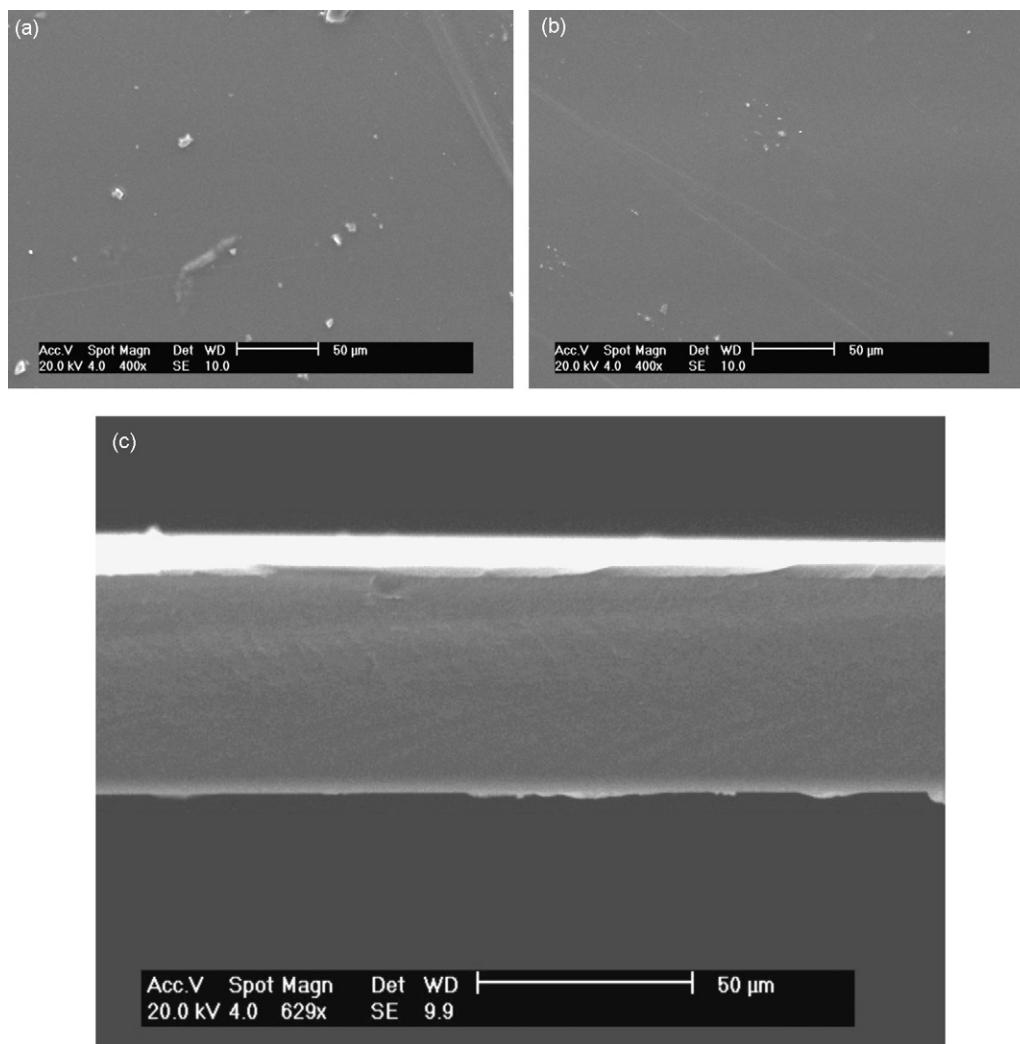


Fig. 6. SEM micrographs for CS membrane: (a) bottom surface; (b) top surface and (c) cross-section.

in the diffusion coefficient of paracetamol. The results also show that, as in the case of pure chitosan membrane, an increase in drug concentration causes a decrease in the permeability coefficient. However, on macroporous membranes, the decrease in permeability coefficient values with concentration are not so drastic and tend to be smaller as the porosity increases (from MPC11 to MPC15). In case of MPC15 the differences in the values of P with concentration practically disappeared. Taking into account the asymmetric structure of the membranes, two distinctive transport mechanisms must be considered: the classic solution-diffusion through the dense polymer layer and the pore-flow mechanism. From the results it can be concluded that the effect of drug concentration is important on the diffusion controlled mechanism but not for the pore-flow one. Considering that the solute sizes are much smaller than the membrane's pore size it can be also postulate that the transport will be determined by the diffusion through the dense layer. Therefore, as the relative dense layer thickness decreases, the permeability tends to decrease. In this case, the same arguments used to explain the variations of permeability with concentration in dense chitosan membranes can be employed here. However, in this case, since the relative dense layer is smaller, it is expected that the macroporous membranes become saturated at lower concentrations and, as a consequence their permeability values tend to achieve a constant value at lower concentrations.

3.4.2. Influence of temperature

The calculated values of permeability, P , for sodium sulfamerazine using different membranes (CS, MPC11, MPC13 and MPC15) and different temperatures are summarized in Table 2. As can be seen from that table, permeability, P , increases with temperature for all four membranes. The influence of temperature on permeability of polysaccharides such as chitosan and/or cellulose, have been studied by several authors [24–27]. Most of them found an increase in permeability with temperature, following the situation classically observed in liquid phase. However, the reverse situation, where the permeability decreases with temperature has also been reported [25]. According to the authors, as the temperature increases, two competing factors may determine solute transport: the network shrinkage, which tends to lower the mass transport and, the thermal activation, which increases the mass transport. The data in Table 2 indicate that, for all analyzed samples, the thermal activation seems to be the primary factor that governs the dependence of P with temperature. This could be due to the fact that the increase of temperature elevates the polymer thermal mobility and consequently leads to the rise of mass transport across the membrane. It is worthwhile to note that for MPC15 membrane, which has the higher porosity, the differences in permeability are not so evident, indicating that the temperature effect is more important on the dense layer than on the porous one.

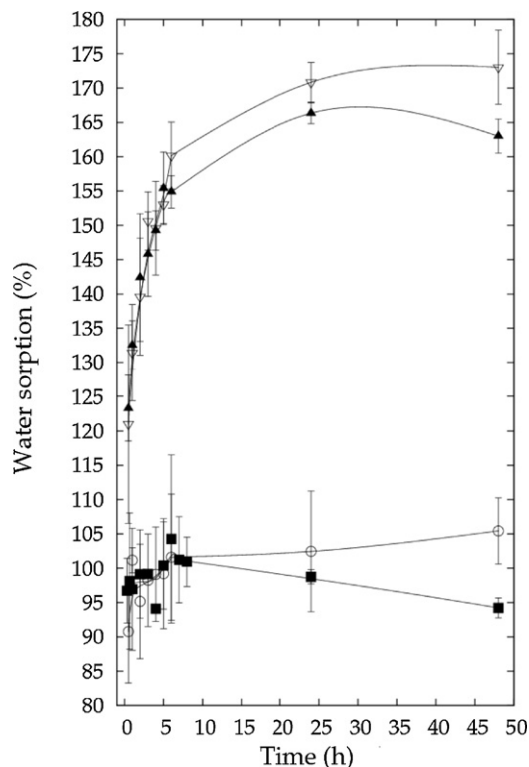


Fig. 7. Water sorption experiments. Full square, pure chitosan, CS. Open circle, MPCSH11. Up triangle, MPCS13. Down triangle, MPCS15.

3.4.3. Influence of solute

In order to examine whether the permeation mechanism differs depending on the drugs species, a series of studies was conducted using two model drugs belonging to the sulfa family. Sulfonamide drugs were the first antimicrobial drugs and paved the way to the antibiotic revolution in medicine in 1932. Many thousands of molecules containing the sulfanilamide structure have been created since its discovery yielding improved formulations with greater effectiveness and low toxicity. Sulfa drugs are still widely used for conditions such as acne and urinary tract infections and are receiving renewed interest for the treatment of infections caused by bacteria resistant to other antibiotics [28]. Besides, new pharmaceutical formulations like hydrogel nanoparticles for controlled release have also been investigated [29]. The choice of the drugs was also based on their similarity in structure and molecular weight as can be seen in Fig. 2. The basic difference between them is that sodium sulfamerazine is under its salt form, while sulfamethoxypyridazine is a non-ionic solute. Fig. 8 shows some typical permeation curves for sodium sulfamerazine. As can be observed, the induction time is negligible and a constant rate stage directly followed. Similar results were obtained for sulfamethoxypyridazine. The absence of induction time is usually attributed to thin membrane thickness [30], while the steady rate of permeation is quite common in membrane controlled release systems, because it is possible to maintain a constant driving force, *i.e.*, concentration difference between donor and receptor side [31]. The calculated values of permeability for sulfamerazine and sulfamethoxypyridazine, respectively, are shown in Tables 2 and 3. As can be observed, sulfamerazine was evaluated over a broad range of concentration while for sulfamethoxypyridazine only the lower concentration was used. The use of higher concentrations for sulfamethoxypyridazine was limited by its low solubility. The results indicate that, although very similar in structure, the drugs present quite different values of permeability. For all kinds of analyzed membranes can be observed

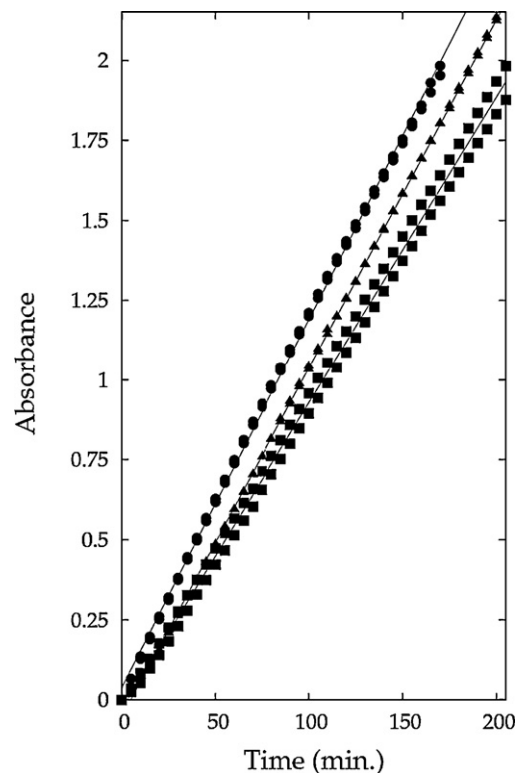


Fig. 8. Absorbance versus time curves for permeation of sodium sulphamerazine on MPCH11 at different temperatures. Full square, 30 °C. Full triangle 35 °C. Full circle 40 °C.

that sulfamethoxypyridazine permeability values are always smaller than the ones calculated for sodium sulfamerazine. The first argument usually used to explain differences in permeability is the presence of solute–membrane interactions. Considering chitosan membranes it is assumed that, at low pH values, the membrane is positively charged and therefore could interact ionically with negatively charged solutes [11]. However, it does not seem to be the case in this work, especially because both drugs have a basic character forming solutions with pH 8.8 (sodium sulfamerazine) and pH 7.8 (sulfamethoxypyridazine). Such values are well above the pK_a value found for chitosan (between 6.2 and 6.5 according to molecular weight and/or deacetylation degree [32]), indicating that the membrane is not under its protonated form and therefore could not ionically interact with the solute.

Apart from the ionic interactions, another factor that could explain the observed differences in permeability for the drugs in question is their water solubility. Sodium sulfamerazine, being a charged drug, is highly water soluble (27.8 g/100 ml). On the other hand, sulfamethoxypyridazine, a non-ionic drug, has a much lower water solubility (0.14 g/100 ml) [33]. As the permeation process is governed by the solute dissolution in the water filled pores (for porous membranes) or the water filled voids, for the dense ones, it is expected that the more water-soluble solute will permeate more easily.

4. Conclusions

The suggested methodology was adequate for obtaining macroporous chitosan membranes with asymmetric structure. SEM observations enable us to conclude that the silica particles were deposited at the bottom surface while the top one was dense and uniform. The images also showed an asymmetric morphology for the three analyzed compositions. By the cross-section images it

was possible to conclude that the 1:5 composition is the limit for obtaining an asymmetric membrane and that higher silica content would yield a symmetric membrane with pores in both surfaces. It was also observed that membrane thickness and porosity were directly influenced by the amount of silica used. The membrane pore size was fairly controlled by silica particle diameter, although some higher pores can be generated by the collapse of adjacent pores. Considering the water sorption capacity it is possible to conclude that the higher porosity membrane have higher water sorption capacity and that swelling works opening the network and promote the permeation of the drugs. All membranes presented an inverse correlation between drug concentration and permeability. However, for MPC515 such a correlation is less evident, indicating that for this membrane the pore-flow mechanism predominates. The permeation increased with temperature for all analyzed membranes although this effect is once again less evident on MPC515 membrane. The differences in permeability observed for the drugs in study were explained in terms of their different solubilities.

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